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Source domain prior-assisted segment anything model for single domain generalization in medical image segmentation

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ABSTRACT

Deep learning-based medical image segmentation models often suffer from performance degradation across domains due to domain discrepancies arising from data collected by various healthcare centers. Recent advancements, particularly the Segment Anything Model (SAM), have shown promising generalization abilities in this field, inspiring the development of several SAM-based approaches to address domain discrepancy issue. Nevertheless, these methods seldom exploit the full potential of the rich source domain knowledge to improve segmentation accuracy in unseen target domains. In this paper, we propose a source domain prior-assisted module for generalizable medical image segmentation based on SAM. Specifically, we store diverse features of the source domain data in a memory bank. When applying the model to the target domain, the features of the target domain first match with the features in the memory bank to obtain invaluable prior information. This strategy enables the model to utilize the prior information from the source domain to adapt to the target domain. We validate the proposed method on two widely used medical image segmentation tasks across multiple domains and achieve state-of-the-art performance.

1. Introduction

Medical image segmentation is a vital task in clinical image analysis, focusing on the semantic interpretation of each pixel. Accurate and reliable automatic segmentation is essential for providing clinicians with timely assistance and advice, thereby enhancing the efficiency of clinical workflows. In the past decade, with the development of deep learning [1–11], medical image segmentation has made significant progress. The pioneer U-Net [12] and its variants [13–15] have achieved satisfactory results by effectively combining shallow texture information and deep semantic information through skip connections. With the success of vision transformers (ViT) [16] in natural image recognition and semantic segmentation, some transformer-based method [17,18] have also been proposed for medical image segmentation. Although, deep learning-based medical image segmentation [1–5,19–21] has achieved satisfactory results when the training and testing data belong to the same source domain. However, due to significant domain shifts [22] between medical images collected by different medical centers or different medical devices, models trained on the training domain (source domain) often experience a significant drop in performance when applied to the unseen domain (target domain). To address this issue, many methods [19,21,23–29] have been proposed to enhance the model's generalization performance across different domains.

Unsupervised domain adaptation (UDA) [20,29–32] leverages labeled source domain data and unlabeled target domain data to optimize the model. Some methods [29,31,32] incorporate unlabeled data during training, using techniques such as adversarial training to learn domain-invariant feature representations, enabling the model to better adapt to target domain data. Some methods [20,30] fine-tune the model with unlabeled target domain data during the testing phase to improve the model's performance in the target domain. Although UDA has achieved good performance, its clinical application is limited because these methods require data from the target domain.

Multi-domain generalization (MDG) [19,21,28,33,34] involves training a model on multiple source domains to learn richer knowledge, thus improving the network's generalization performance on the target domain. However, due to privacy concerns associated with medical images, obtaining data from multiple source domains or the target domain is not easily achievable, which limits the practical application of such methods.

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Therefore, a more practical method, single domain generalization (SDG) has been proposed which is trained on a single source domain. However, a single domain lacks diverse training samples, making the model prone to overfitting on the source domain data, which results in poor model generalization. Therefore, some methods [24–26,35] have been proposed to enhance the model's generalization performance through data augmentation on input data or features, or by designing specific network structures.

Recently, vision foundation models have made significant breakthroughs in visual tasks. The Segment Anything Model (SAM) [36], trained on a vast number of natural images, has demonstrated powerful segmentation capabilities for image segmentation tasks. Currently, many research outcomes [37–41] indicate that directly applying this pre-trained model to medical image segmentation tasks still yields satisfactory results. Some methods [23,37–42] also fine-tune SAM to make it better suited for medical image segmentation tasks. There are also some studies [23,42] build upon SAM to further enhance the generalization capabilities of the model in medical image segmentation tasks.

In this paper, considering that the model can better fit the training data (i.e., source domain data), it demonstrates that the model can extract more effective and richer source domain information. If the prior information from the source domain data can be utilized during the model's testing phase, it can help improve the model's generalization performance. Based on this, we propose adding a source domain prompt module to the SAM framework. Specifically, we store the features of the source domain data into a memory bank. When applying the model to the target domain, the features of the target domain strategically match with those preserved in the memory bank to capture valuable source domain prior information, and then the newly obtained features serve as the prompt for the mask decoder to assist segmentation. Furthermore, taking into account that handling variations in intensity values across different domains poses a critical challenge in medical image segmentation, we propose to select some source domain data with the greatest intensity differences to update the memory bank. This approach significantly enhances the network's adaptability within the target domain. It is worth noting that, unlike the standard SAM which utilizes a fixed architecture often unable to adapt effectively to unseen domains, our method incorporates Low-Rank Adaptation (LoRA) layers. These layers allow for dynamic adjustments to the feature space, thereby enhancing the model's flexibility and adaptability to unseen domains.

The main contributions of this work are as follows:

1) We propose to utilize source domain prior information as prompts to enhance the model's generalization performance. To achieve this, we develop a Source Domain Prior-Assisted Module (SDPA), consisting of a memory bank to store the source domain prior and a memory read component for strategically matching these prior to extracting valuable information.
2) We propose to select source domain images with the largest intensity differences for updating the memory bank. This strategy provides a richer source domain prior, thereby addressing the challenge of significant intensity distribution differences between different domains in medical image data.
3) We implement our method based on the SAM and incorporate the trainable LoRA layer into the image encoder of SAM. This approach not only retains the original SAM's capability to extract features but also introduces flexibility, enabling the model to adapt to specific domains.

The rest of this paper is organized as follows. Some related works are reviewed in Section 2, followed by the detail of the proposed method in Section 3. We present extensive experimental results in Section 4. We then discuss these findings in Section 5. Finally, we give a conclusion in Section 6.

2. Related work

2.1. Domain generalization

Medical imaging data collected from different centers or medical devices exhibit a domain gap. This results in that deep neural network models trained on one domain may experience a significant performance drop when applied to an unseen domain. As a result, the model may not be suitable for clinical tasks. To address this issue, various methods have been proposed to enhance the model's generalization capabilities. This type of problem is referred to as domain generalization, which can be roughly divided into three categories: unsupervised domain adaptation (UDA), multi-domain generalization (MDG) and single domain generalization (SDG).

UDA aims to address domain shift issues by incorporating unlabeled target domain data during the training process. Some methods [29,43–45] align the source and target domains at the input data or feature level, enabling the model to learn domain-invariant characteristics and thus improve its segmentation performance on the target domain. For instance, Chen et al. [29] proposes enabling the model to learn domain-invariant representations at both the input image and intermediate feature levels while employing adversarial learning to align the boundaries and regions obtained from the source and target domains. UPL-SFDA [31] introduces an effective source-free domain adaptation method for medical image segmentation that enhances the quality of pseudo labels. Some methods [20,46] fine-tune the source domain model during testing, transferring knowledge learned in the source domain to the target domain. Karani et al. [20] design a test-time adaptation method comprising a shallow image normalization network and a segmentation network for normalized images. DIGA [46] propose a backward-free method for test-time domain adaptation with low computational cost. Although previous UDA methods have achieved impressive results in generalization tasks, obtaining target domain data during the training process is challenging due to medical data privacy issues. Additionally, fine-tuning the model during the testing phase inevitably requires local clients to possess more computational resources.

MDG utilizes data from multiple source domains for training, enabling the model to acquire richer information for improved performance in the target domain. Some studies improve the generalization of models by establishing relationships between multiple source domains. For example, DCAC [28] establishes relationships between multiple source domains and the target domain based on domain-discriminative information obtained from the encoder, thus facilitating the model's better adaptation to the target domain data. MDViT [33] proposes mitigating data scarcity issues by adaptively leveraging knowledge from multiple small-scale data domains and integrating a distillation strategy to strengthen cross-domain representation learning. There are also methods to enrich the feature representation of the data so that the model is better adapted to the target domain. Gao et al. [47] presents a decentralized distributed multi-source domain training framework achieved by addressing label bias and regularizing features. TriD [19] proposes random sampling from a uniform distribution, constructs a novel search space, and employs a style mixing strategy to obtain a model with better generalization. However, MDG requires multiple annotated training domain datasets, and pixel-level annotations necessitate specialized expertise and significant expenditure. Additionally, collecting data from various centers is challenging due to privacy concerns.

SDG is trained on a single source domain. For enhancing the network's generalization ability on unseen domains. Some methods propose effective data augmentation methods to increase the distribution of training samples. Such as, MaxStyle [48] proposes the addition of data augmentation in the feature space, which involves feature reconstruction of input images and using these reconstructed images as additional training data for the network. Su et al. [25] propose two new

augmentation strategies to effectively improve the generalization of the network. Some research design network structures to learn a more diverse set of information. For instance, C^2SDG [26] decouples style and structural representations, subsequently reducing the impact of style-sensitive channels. SDG is a more practical yet more challenging approach. Although existing methods have achieved commendable generalization effects, they have not effectively utilized the prior knowledge from the source domain during the testing phase.

2.2. SAM for medical image segmentation

With the widespread attention to Segment Anything Model (SAM) [36] in natural image segmentation tasks, numerous recent works, built upon SAM, have been proposed for medical image segmentation, yielding satisfactory results. SAM-Med2D [49] conducted extensive experiments to thoroughly validate and analyze SAM's performance in medical image segmentation tasks. SkinSAM [50] fine-tunes SAM for skin cancer segmentation tasks and discusses the impact of prompts on the network's performance. SAMed [38] applies the low-rank-based finetuning strategy in SAM's image encoder to optimize the network for medical image segmentation. MSA [37], which applies the adapter technique widely used in natural language processing to SAM, achieves an efficient and general medical segmentation method. MedSAM [51] is trained on over one million medical images, obtaining a foundational medical model with significant potential for medical diagnostic assistance. DeSAM [23], based on SAM, introduces a skip-connection structure similar to U-Net, which combines shallow features from SAM's image encoder with deep features from the Mask decoder, enhancing the model's generalization performance. Although these large models have impressive learning capabilities, their generalization ability in medical tasks still needs improvement due to the lack of learning from specialized medical data.

In this work, focusing on the practical application of the SDG method, we propose a source domain-prior prompt SDG model for medical image segmentation. Concretely, we propose storing the features of the source domain data. When applying the model to segment data in the target domain, based on the features of the target domain, we extract valuable information from the stored source domain features to serve as the prompt for the mask decoder. Our goal is to enhance the model's ability to generalize within the target domain.

3. Methodology

3.1. Overview

The overall pipeline is shown in Fig. 1. It comprises a standard SAM model, LoRA layers, and a source domain prior-assisted (SDPA) module. Following SAMed [38], we freeze all parameters of the image encoder in the original SAM. A trainable LoRA is inserted as a branch between each transformer block. It first compresses the transformer features into a low-rank space and then re-projects the compressed features to align with the output feature channels of the frozen transformer block. The default prompt in SAM is fine-tuned for the prompt encoder. The mask decoder of SAM can be divided into a lightweight transformer decoder and a segmentation head decoder, with all its parameters fine-tuned during the model training process. The source domain prior-assisted (SDPA) module is designed to enhance the model's generalization capabilities. Specifically, we adopt a memory bank to store the features with prior information of the source domain data. Put it simply, we select M reference data with the diverse intensity in the source domain. Then, at the beginning of each epoch of the network iteration training, these selected reference data are passed through the image encoder to obtain corresponding features, which are used to update the memory bank. In the subsequent iteration training process, the current training image embedding interacts with the prior features stored in the memory bank in the SDPA module. The resulting feature, rich in source domain information, serves as input to the prompt encoder.

3.2. Reference data selection

In medical image, there is a significant variation in intensity values, especially for images from different domains. Therefore, in order to train the network more effectively and obtain a more representative reference set of source domain data, we select M reference images with large intensity differences in the source domain for constructing the memory bank. Concretely, we first calculate the mean intensity μ_k of each image x_k:

$$\mu_k = \frac{1}{|\Omega|}\sum_{p\in\Omega} x_k(p), \tag{1}$$

where $|\Omega|$ denotes the number of pixels in the image x_k. Then, we sort all

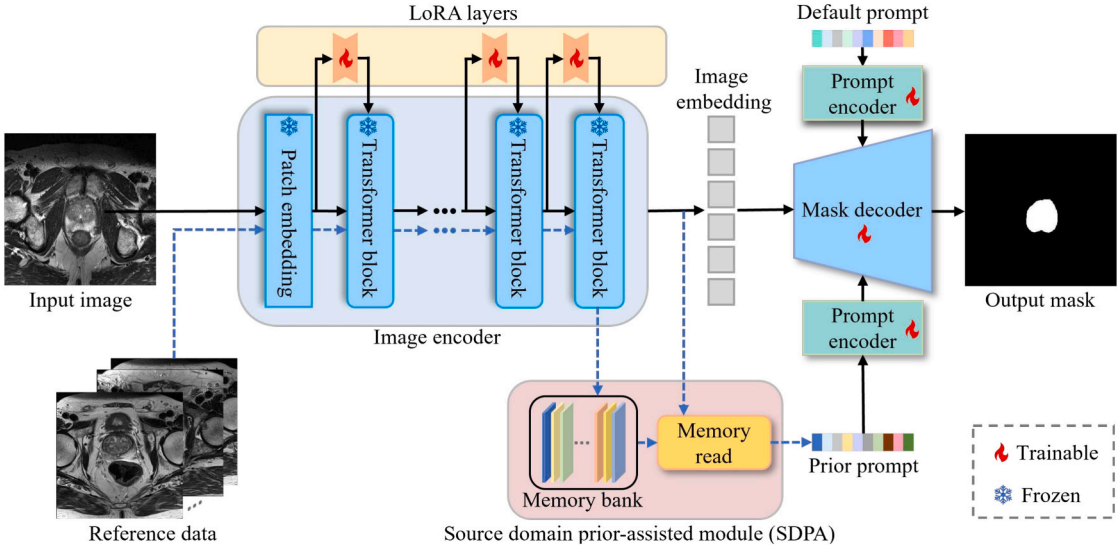


Fig. 1. The pipeline of the proposed method. This framework is built on SAM, freezing its original image encoder's transformer blocks and incorporating trainable LoRA layers for effectively extracting task-specific features. Additionally, we add a source domain prior-assisted module (SDPA) to provide source domain-related prompts to improve the model's performance on unseen target domains. The blue dashed arrow indicates no gradient backpropagation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

images in the source domain with respect to increasing order of mean intensity μ. We select the M images from the sorted source domain images with index $[0, N/M, 2^*N/M, 3^*N/M, \ldots]$ to construct the memory bank, where N is the total number of images in the source domain. Such M images have large variances in intensity distribution and serve as the reference data.

3.3. Source domain prior-assisted module (SDPA)

As shown in Fig. 1, the source domain prior-assisted module (SDPA) consists of a memory bank that stores source domain prior information and a memory read module that performs strategic matching based on the input data.

Memory bank update: We build a memory bank to store the feature of the reference data. Specifically, the reference image is initially processed through the image encoder to acquire image embeddings, which are then stored in the memory bank. At the beginning of each epoch, the memory bank is updated with the latest features of the reference images.

Memory read: As shown in Fig. 2, in the memory read operation, we first calculate the pixel-by-pixel similarity $S_l \in \mathbb{R}^{hw\times hw}$ between the feature $F_c \in \mathbb{R}^{c\times h\times w}$ of the current training image and the l-th feature $F_l^m \in \mathbb{R}^{c\times h\times w}$ of the source domain data stored in the memory bank. Let s_{ij} denote the similarity value corresponding to a pixel in the matrix S_l. Let f_i^c represent the i-th pixel in the F_c and f_j^m represent the j-th pixel in the F_l^m. The similarity s_{ij} between f_i^c and f_j^m can be computed as follows:

$$s_{ij} = \frac{f_i^c \left(f_i^m\right)^{\mathrm{T}}}{\left\|f_i^c\right\|_2 \left\|f_j^m\right\|_2}, \tag{2}$$

where $(\cdot)^{\mathrm{T}}$ indicates the transpose operation.

Then, we can obtain the prior feature $F_l^p \in \mathbb{R}^{c\times h\times w}$ by extracting the features in memory bank that are useful for the current training data based on the similarity matrix:

$$F_l^p = S_l \otimes F_l^m, \tag{3}$$

where $\otimes$ denotes matrix multiplication.

Finally, we sum the prior features obtained from the above operations over all the features in the memory bank to obtain the final prompt feature F_p:

$$F_p = \sum_{l\in M} F_l^p. \tag{4}$$

This prompt feature F_p is then passed through a prompt encoder that has the same structure as the dense prompt encoder in SAM.

Fig. 3 presents the t-SNE visualization of the features in the memory bank, image embeddings, and prior prompt features. It can be observed that the prior prompts, refined through the SDPA module, successfully incorporate prior information from the source domain stored in the memory bank. Consequently, they provide the model with source domain-related and more valuable prompts, thereby enhancing the model's generalization performance.

4. Experiments

4.1. Datasets and evaluation protocols

We adopt the prostate and joint optic cup (OC) and optic disc (OD) segmentation datasets RIGA+ for the validation of the proposed method. The details of the corresponding segmentation tasks and involved datasets are given in the following:

Prostate segmentation in MRI images: The prostate data consists of 6 domains [53–55]: ISBI, ISBI_1.5, I2CVB, UCL, BIDMC and HK. Based on the previous work [56], we preprocess the data, retaining only the slices that contain the prostate region and resizing them to a uniform size of 384 × 384. We utilize the Dice score (Dice) metric, which is a widely used metric in medical image segmentation for 3D volume prostate segmentation, to assess the 2D segmentation capabilities of the network.

OC/OD segmentation in color fundus image: RIGA+ [57,58] consists of 5 domains: Magrabia, BinRushed, BASE1, BASE2, and BASE3. Following [26], we designate the first two datasets, Magrabia and BinRushed, respectively, as the training set, while the subsequent three datasets serve as the test set to evaluate the model's generalization capabilities. We also adopt Dice as the evaluation metric.

4.2. Implementation details

We reproduced some state-of-the-art methods including SAMed [38], DeSAM [23], and CCSDG [26] for comparison with our methods. We chose SAMed [38] as the baseline and built our method on it. We used AdamW as the optimizer and maintained the same training strategy as SAMed, including warmup, learning rate, rank of LoRA, number of iterations, and other settings. Some hyperparameters of the model are detailed in Table 1 We employed various data augmentations, just as in [19], including brightness transform, gamma transform, Gaussian noise, and so on. The amount of source domain data M used to update the memory bank is set to 50 in all experiments. All the experiments are conducted using the Pytorch framework on a workstation with an NVIDIA GeForce RTX 3090 GPU.

4.3. Experimental results

Prostate segmentation: Initially, we assess the generalization capability of the proposed method on the prostate segmentation task. As depicted in Table 2, the methods based on SAM, DeSAM and SAMed, demonstrate a clear improvement in generalization compared to

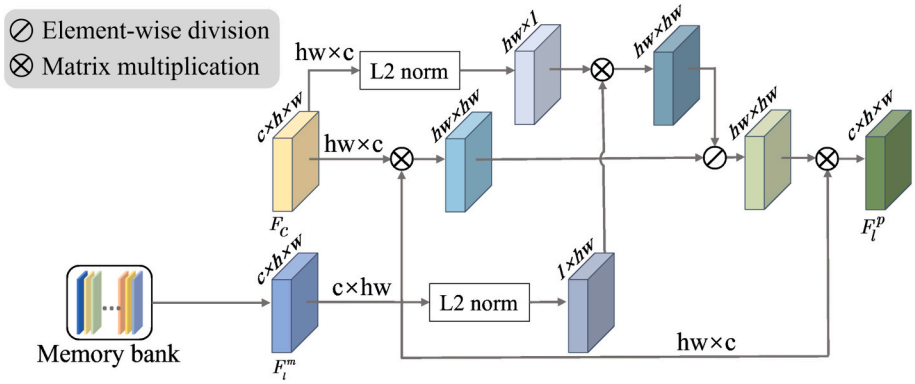
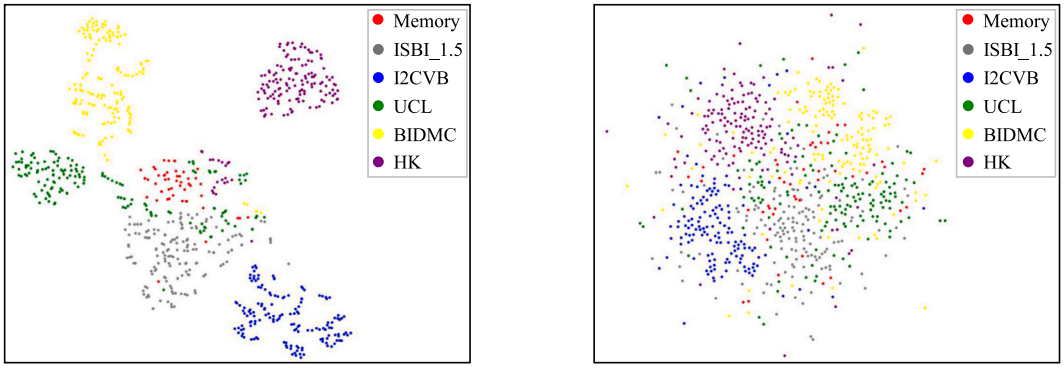


Fig. 2. Detailed implementation of the memory read operation using basic tensor operations as described in Section 3.3.



(a) t-SNE visualization of image embedding (b) t-SNE visualization of prior prompt

Fig. 3. Qualitative t-SNE [52] results of the image embedding (a) and the prior prompt feature (b). The red dots represent the t-SNE visualization of the feature stored in the memory bank. The model is trained on the ISBI domain of prostate segmentation. The other five domains serve as test domains. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Hyper-parameters of the model. *M* represents the number of source domain reference data used to update the memory bank.

SAM version	Rank of LoRA	No. of LoRA layers	*M*
vit_b	4	12	50

methods based on convolutional networks, outperforming the latest methods CSDG and CCSDG by more than 10% in Dice score. By implementing our method based on SAMed, we can further enhance the model's generalization capabilities. Compared to the SOTA methods, our approach consistently delivers superior results. Particularly, compared to the baseline, our method achieved significant improvements, with an average increase in the Dice score of 2.53%. Moreover, when compared to the recent method for solving the domain generalization problem based on vision foundation models SAM, DeSAM [23], our method also shows a notable enhancement. Specifically, compared with DeSAM in whole box mode, the proposed method achieves an improvement of 3.40% Dice score. The proposed method also outperforms DeSAM in grid box mode by 2.92% Dice score. Some qualitative results are illustrated in Fig. 4, our method achieves anatomically correct segmentation results compared to other methods.

OC/OD segmentation: We then validate our method on the RIGA+ dataset. The quantitative results using Magrabia as the source domain training data are presented in Table 3. It is observed that the generalization ability of DeSAM is not as strong as that of other convolution-based methods, and SAMed also fails to demonstrate satisfactory generalization performance. This suggests that SAM-based methods may not consistently achieve superior generalization across all medical tasks, indicating a need for further optimization tailored to specific tasks. Compared to existing methods, our method achieves the best results. Precisely, the proposed method improves the baseline SAM model by 2.17% Dice score on the optic cup (OC) and 0.18% Dice score on optic disc (OD). Compared to DeSAM, our method exhibits significant improvements in both OC and OD. (with a Dice score increase of over 10%). In Table 4, the quantitative results using BinRushed as the source domain for training data are presented. Our method delivers results that are on par with the SOTA method CCSDG. Our method exhibits 4.07% improvement in Dice score for OC and 1.06% improvement for OD compared to the baseline. There are also important enhancements compared to DeSAM.

Our method yields more precise segmentation results in the target domain compared to several recent methods, as illustrated in Fig. 4, further demonstrating the good generalization capability of our method.

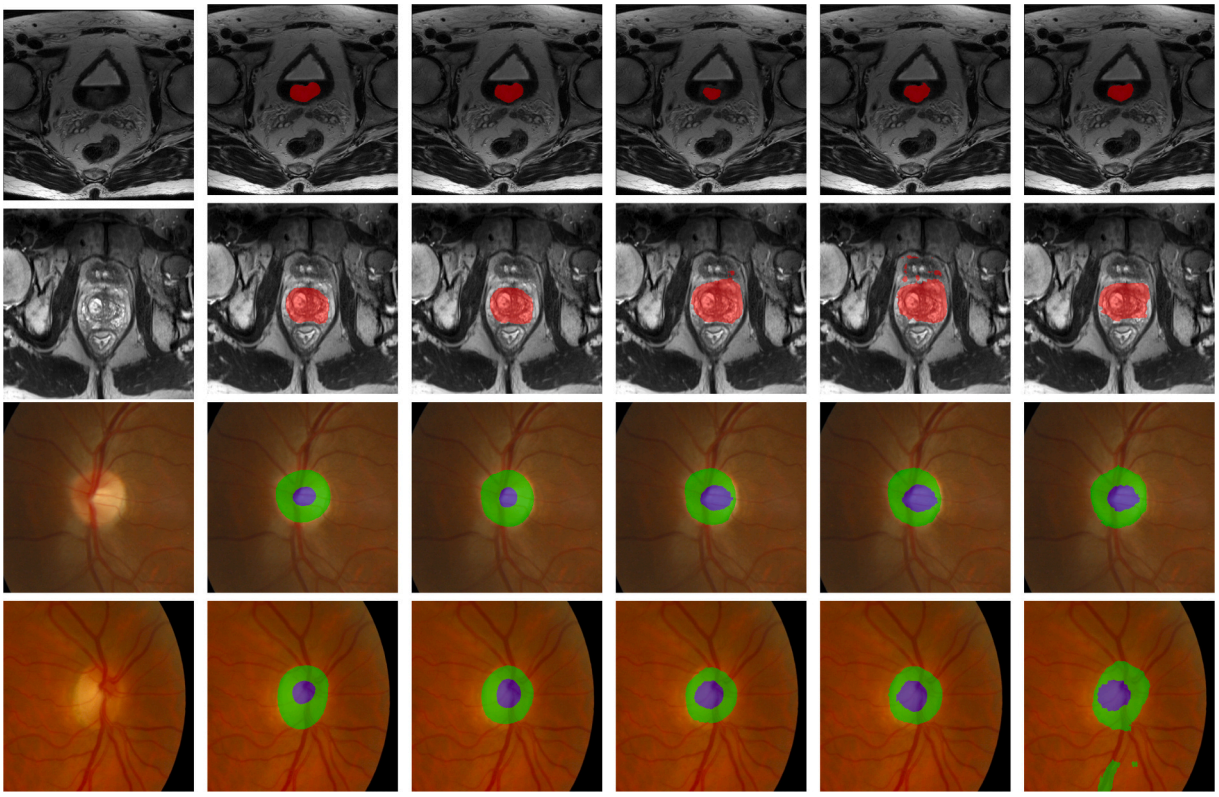
4.4. Ablation study

We conduct three types of ablation studies on prostate segmentation: 1) Investigating the hyper-parameter *M* utilized in constructing the memory bank; 2) Examining the effects of the reference data selection strategy; 3) Assessing the impact of the number of LoRA layers on model generalization.

Ablation study on the hyper-parameter *M*: The number of source domain reference images used to update the memory bank directly influences the network's generalization ability. Intuitively, the more reference images used, the richer the prior information about the source domain, leading to better network generalization. However, this also results in increased memory usage. We set the value of *M* to 0, 25, 50, 75, and 100, where $M = 0$ represents the baseline model. As depicted in Fig. 5, our method consistently outperforms the baseline model. This also confirms our hypothesis that using more reference data can lead to better results. This demonstrates that our method effectively enhances

Table 2
Quantitative comparison between the proposed method and several competing methods on prostate segmentation. DeSAM [23] (w) and DeSAM [23] (g) indicate the whole box mode and the grid points mode, respectively. We build the proposed method upon SAMed [38].

Method	ISBI to rest	ISBI_1.5 to rest	I2CVB to rest	UCL to rest	BIDMC to rest	HK to rest	Average
AdvNoise [15]	72.15	63.26	30.81	40.1	48.07	60.1	52.42
AdvBias [15]	77.45	62.12	51.09	70.20	51.12	50.69	60.45
RandConv [59]	75.52	57.23	44.21	61.27	49.98	54.21	57.07
MixStyle [60]	73.04	59.29	43.00	62.17	53.12	50.03	56.78
MaxStyle [48]	81.25	70.27	62.09	58.18	70.04	67.77	68.27
CSDG [24]	80.72	68.00	59.78	72.40	68.67	70.78	70.06
CCSDG [26]	80.62	69.52	65.18	67.89	58.99	63.27	67.58
DeSAM [23] (W)	83.10±2.14	77.54±2.56	65.93±3.23	83.02±1.62	76.95±4.37	79.32±1.96	77.64
DeSAM [23] (G)	81.24±2.56	79.45±3.07	62.20±2.21	82.58±1.53	81.46±3.96	**81.77±2.42**	78.12
SAMed [38]	80.42±2.89	81.44±3.11	66.75±4.21	82.09±1.56	80.19±5.89	80.17±2.64	78.51
Ours	**85.76±1.23**	**83.12±2.12**	**68.74±2.55**	**83.74±1.09**	**83.74±2.37**	81.16±2.04	**81.04**



(a) Image (b) GT (c) Ours (d) SAMed (e) CCSDG (f) DeSAM

Fig. 4. Visualization of some segmentation results predicted by our method and several competitive methods on prostate segmentation (Top two rows) and OC/OD segmentation (Bottom two rows).

Table 3
Quantitative evaluation of OC/OD segmentation using Magrabia as source domain.

Method	BASE1		BASE2		BASE3		Average	
	OD	OC	OD	OC	OD	OC	OD	OC
BigAug [61]	92.32±0.13	79.68±0.38	88.24±0.82	76.69±0.37	91.35±0.14	81.43±0.78	90.64	79.27
CISDG [24]	89.67±0.76	75.39±3.22	87.97±1.04	76.44±0.48	89.91±0.64	81.35±2.81	89.18	77.73
ADS [27]	90.75±2.42	77.78±4.23	90.37±2.07	79.60±0.34	90.34±2.93	79.99±4.02	90.48	79.12
MaxStyle [48]	91.63±0.1	78.74±1.95	90.61±0.45	80.1±0.90	91.22±0.07	81.90±1.14	91.15	80.25
SLAug [48]	93.08±0.17	80.70±0.35	92.70±0.1	80.1±0.43	92.23±0.16	80.89±1.14	92.67	80.58
CCSDG [26]	94.78±0.03	84.94±0.36	95.16±0.09	85.68±0.28	95.00±0.09	85.90±0.29	94.98	85.53
DeSAM [23] (W)	76.14±0.04	63.31±0.26	79.97±0.10	72.84±0.26	75.87±0.11	69.20±0.25	77.33	68.45
DeSAM [23] (G)	83.37±0.08	69.81±0.31	85.27±0.13	78.29±0.31	82.08±0.14	74.62±0.29	83.57	74.24
SAMed [38]	95.41±0.10	**85.26±0.38**	95.36±0.13	84.25±0.27	95.38±0.1	84.76±0.28	95.38	84.76
Ours	**95.42±0.05**	84.54±0.29	**95.51±0.10**	**88.39±0.25**	**95.74±0.08**	**87.85±0.23**	**95.56**	**86.93**

Table 4
Quantitative evaluation of OC/OD segmentation using BinRushedas source domain.

Method	BASE1		BASE2		BASE3		Average	
	OD	OC	OD	OC	OD	OC	OD	OC
BigAug [61]	94.01±0.34	81.51±0.58	85.81±0.68	71.12±1.64	92.19±0.51	79.75±1.44	90.67	77.46
CISDG [24]	93.56±0.13	81.00±1.01	94.38±0.23	83.79±0.58	93.87±0.03	83.75±0.89	93.93	82.85
ADS [27]	94.07±0.29	79.60±5.06	94.29±0.38	81.17±3.72	93.64±0.28	81.08±4.97	94.00	80.62
MaxStyle [48]	94.28±0.14	82.61±0.67	86.65±0.76	74.71±2.07	92.36±0.39	82.33±1.24	91.09	79.88
SLAug [48]	95.28±0.1	83.31±1.10	95.49±0.16	81.36±2.51	95.57±0.06	84.38±1.39	95.45	83.02
CCSDG [26]	95.73±0.08	86.13±0.07	95.73±0.09	**86.82±0.58**	95.45±0.04	86.77±0.19	95.64	**86.57**
DeSAM [23] (W)	85.20±0.07	69.49±0.06	88.82±0.09	79.16±0.52	85.41±0.07	74.98±0.28	86.48	74.54
DeSAM [23] (G)	83.43±0.10	70.68±0.13	84.96±0.14	77.87±0.72	79.84±0.09	72.23±0.30	82.74	73.59
SAMed [38]	95.28±0.07	84.24±0.10	94.11±0.1	80.21±0.64	94.84±0.08	82.60±0.32	94.74	82.35
Ours	**95.84±0.08**	**86.19±0.05**	**95.81±0.07**	85.60±0.49	**95.75±0.02**	**87.46±0.13**	**95.80**	86.42

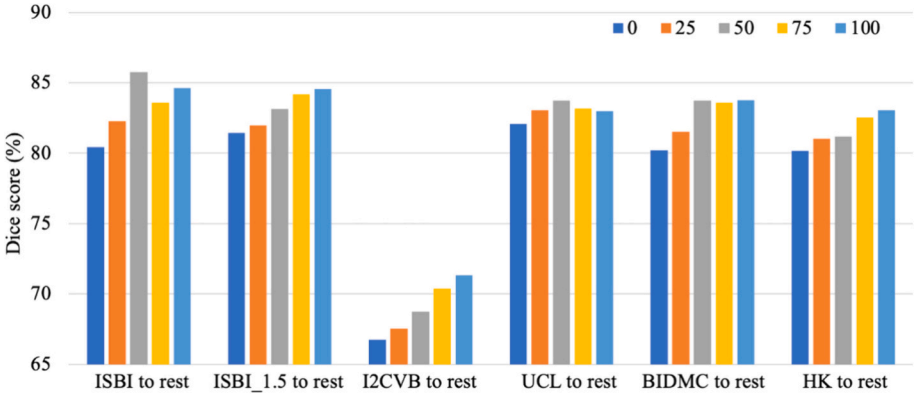


Fig. 5. The effect of selecting different numbers M of reference data to construct the memory bank.

the network's generalization capability.

Ablation study on the effect of the selection strategy for reference data: In our experiments, we propose a strategy of selecting 50 source domain samples with the largest intensity differences to update the memory bank. An intuitive alternative would be to randomly select 50 source domain samples for the update. To validate the effectiveness of our proposed strategy, we conducted ablation experiments for comparison. As depicted in Table 5, our method outperforms the method of randomly selecting reference data, indicating that selecting reference data with a large variation of intensity distribution helps improve the model generalization performance.

Ablation study on the impact of the number of LoRA layers: By inserting trainable LoRA layers as a branch between each transformer block, the SAM model pre-trained on natural images can better adapt to medical imaging contexts. To assess the impact of the number of LoRA layers on model generalization, we conducted experiments based on SAM and our method. As shown in Fig. 6, the model's generalization performance improves with an increase in LoRA layers. This is an expected phenomenon since more LoRA layers enable the image encoder to better adapt to specific medical imaging scenarios and extract more effective features. Notably, our proposed method consistently outperforms SAM.

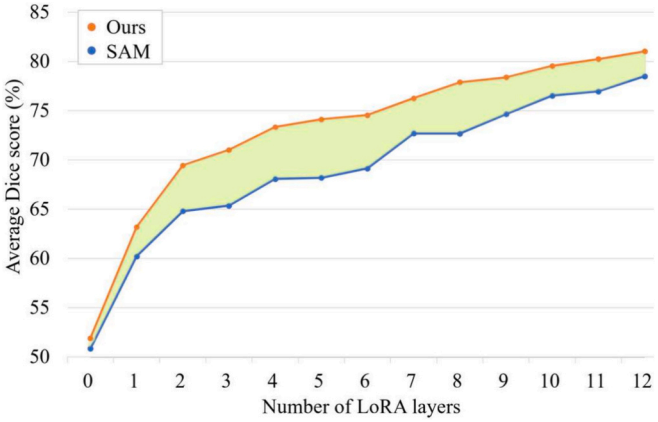


Fig. 6. Effect of varying the number of LoRA layers. When the number of LoRA layers is zero, only the mask decoder and prompt encoder are fine-tuned.

5. Discussion

Medical image segmentation plays a crucial role in clinical practice. With the advancement of deep learning, numerous approaches have achieved remarkable results. However, due to the domain shift that exists in data collected by different medical centers or medical devices, the performance of trained models often significantly declines when applied to unseen domains. This issue has limited the clinical application of the trained model.

Various methods have been proposed to address this problem, known as the domain generalization problem, which can be broadly categorized into three types: unsupervised domain adaptation (UDA), multi-domain generalization (MDG), and single domain generalization (SDG). Recently, with the impressive performance of vision foundation models in natural image tasks, an increasing number of methods have applied these models to medical image segmentation tasks, achieving satisfactory results.

In this work, we propose a single domain generalization model for medical image segmentation based on the vision foundation model, Segment Anything Model (SAM), with a source domain prior prompt. Specifically, we introduce a storage module in the network to preserve features from the source domain data. When the model is applied to segment target domain data, the features of the target domain are strategically matched with those in the storage module to extract valuable information from the stored source domain features as a prompt for the mask decoder. Our goal is to enhance the model's generalization ability in the target domain. We have validated our proposed method on two widely-used medical image segmentation tasks involving multiple domains, achieving state-of-the-art performance.

Although our method has achieved satisfactory generalization effects for SDG, it also has some limitations. For instance, the addition of a memory bank to store features of source domain reference images, along with the memory read operations, inevitably requires additional computational resources. Furthermore, the reference images used to update the memory bank are selected simply based on the mean intensity values of the images. In future work, we will continue to explore ways to reduce the consumption of computational resources and further investigate how to select more representative reference images.

Table 5
Quantitative results of different selection strategies for reference data on prostate segmentation.

Selection strategy	Average Dice score
Random	80.42
Ours	**81.04**

6. Conclusion

In the field of medical image segmentation, addressing domain discrepancy between data collected from different healthcare centers or devices remains a pivotal challenge. Although current vision foundation models, such as SAM, have made significant progress in natural imaging tasks, they still face challenges when addressing the domain discrepancy issue in medical image segmentation tasks. In this paper, we propose to

leverage the rich prior information of the source domain by using a feature storage memory bank. By strategically matching this with the target domain, the model demonstrates enhanced adaptability, offering effective support for more accurate and reliable medical image segmentation. Extensive experimental results on two medical image segmentation tasks demonstrate the effectiveness of our method.

CRediT authorship contribution statement

Wenhui Dong: Writing – review & editing. **Bo Du:** Supervision. **Yongchao Xu:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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